

EAS 6134: Inverse Methods in EAS

Linear Algebra Review

21 January 2026

A Magnetic Example

- Forward problem - predict magnetic anomaly data B from seamount magnetization vector M
- Inverse problem - find magnetization M from measurements B

Bathymetry Model

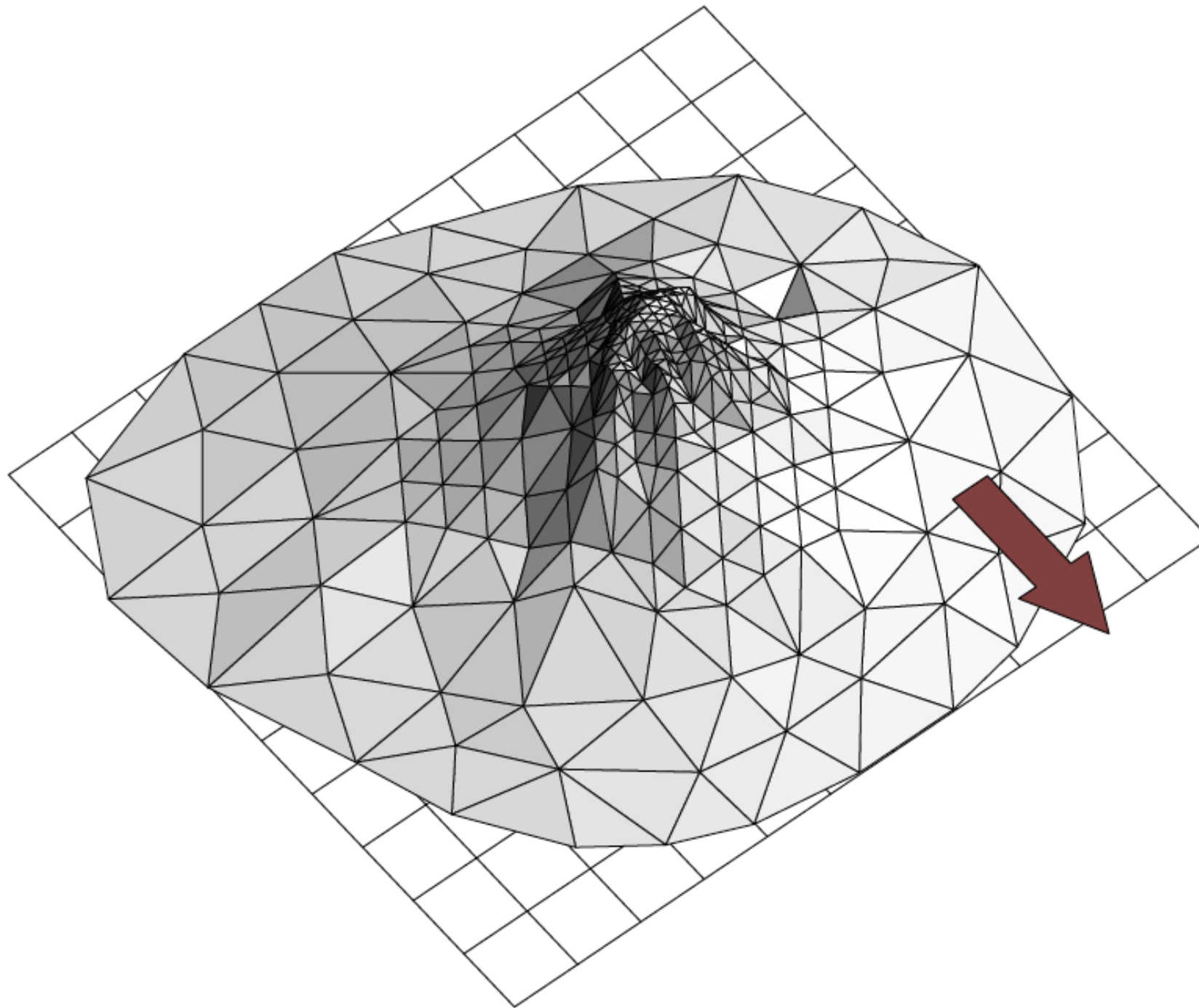


Figure 2.1: Model bathymetry of seamount LR148.8W (48.2°S, 148.8° W). Each square in the base is 5 km on a side and the arrow points north.

Ship Track and Magnetic Anomaly

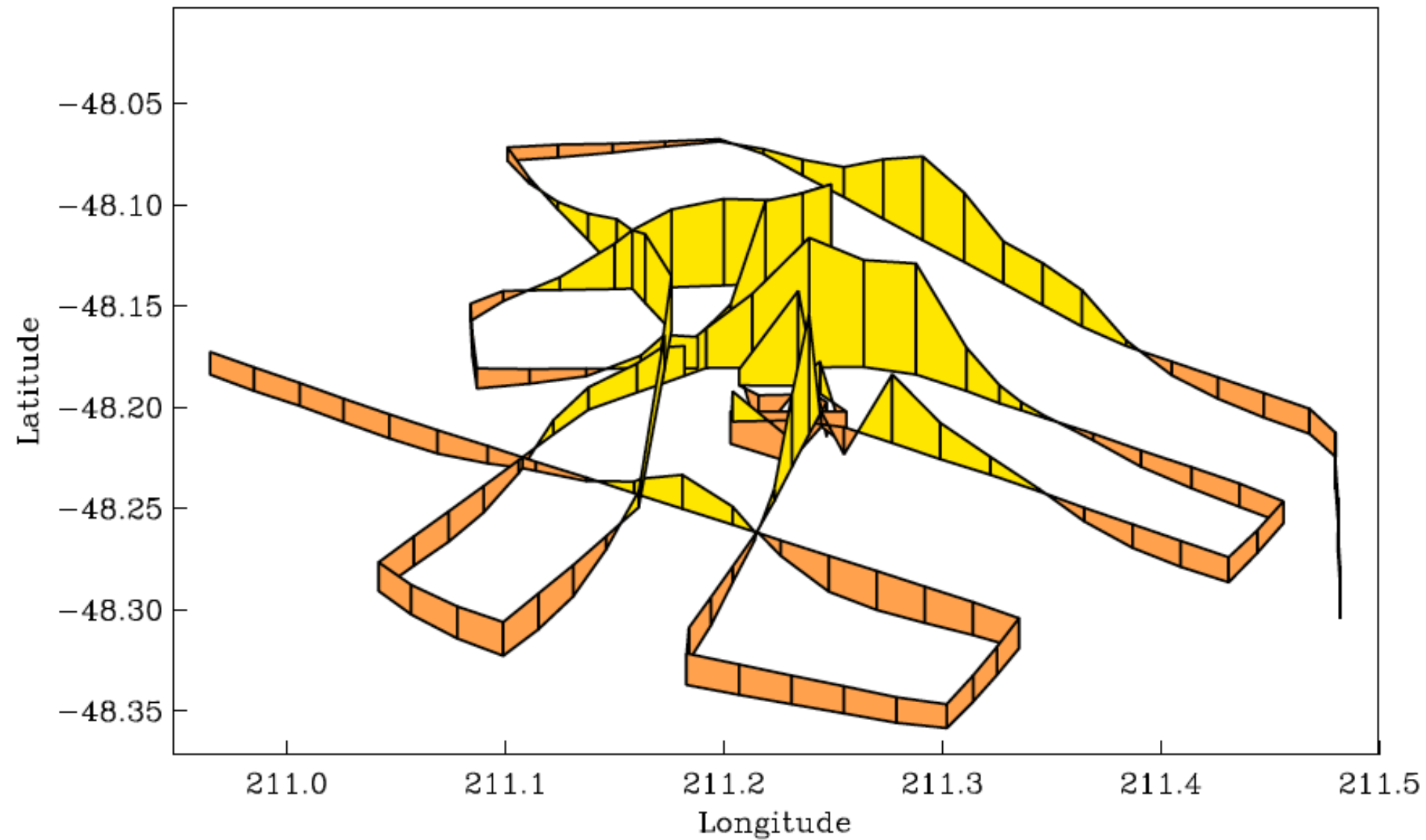


Figure 2.2: Ship track and magnetic anomaly over the seamount LR148.8W.

Forward Problem

$$\Delta \mathbf{B}(\mathbf{r}) = \int_V \mathbf{G}(\mathbf{s}, \mathbf{r}) \cdot \mathbf{M}(\mathbf{s}) d^3\mathbf{s} \quad (1)$$

where $\mathbf{M}$ is the magnetization vector at the point $\mathbf{s}$ within the seamount V and the vector valued function $\mathbf{G}$ is given by

$$\mathbf{G}(\mathbf{s}, \mathbf{r}) = \frac{\mu_0}{4\pi} \hat{\mathbf{B}}_0 \cdot \nabla \nabla \frac{1}{|\mathbf{r} - \mathbf{s}|} \quad (2)$$

where ∇ acts on $\mathbf{s}$, and $\hat{\mathbf{B}}_0$ is a known constant unit vector and $\mu_0 = 4\pi \times 10^{-7} \text{Hm}^{-1} = 100 \text{nT m A}^{-1}$ the permeability of free space in SI units. All the grad operators here act on the coordinate $\mathbf{s}$. Equations (1)-(2) just state that the observed field at $\mathbf{r}$ is the sum of the fields from all the elementary dipoles within V . If we knew $\mathbf{M}$, which is just the density of dipole moments, we could compute $\Delta \mathbf{B}$ from (1) and (2), which means the forward problem has been solved.

What equations would be needed to evaluate your favorite forward problem?

no uniqueness theorem for this inverse problem, even when perfect data like these are available. We demonstrate the nonuniqueness with some simple vector calculus. First rewrite (1) as

$$\Delta \mathbf{B}(\mathbf{r}) = \int_V \nabla g \cdot \mathbf{M}(\mathbf{s}) d^3 \mathbf{s} \quad (3)$$

where

$$g(\mathbf{s}, \mathbf{r}) = \frac{\mu_0}{4\pi} \hat{\mathbf{B}}_0 \cdot \nabla \frac{1}{|\mathbf{r} - \mathbf{s}|} \quad (4)$$

which is OK because $\hat{\mathbf{B}}_0$ is constant. Next consider a magnetization vector inside V given by $\mathbf{m} = \nabla f$ where f is some smooth function. Then (3) becomes

$$\Delta \mathbf{B}(\mathbf{r}) = \int_V \nabla g \cdot \nabla f d^3 \mathbf{s} \quad (5)$$

$$= \int_V [\nabla \cdot (f \nabla g) - f \nabla^2 g] d^3 \mathbf{s} \quad (6)$$

where I have used the very useful vector identity: $\nabla \cdot (f \mathbf{V}) = \nabla f \cdot \mathbf{V} + f \nabla \cdot \mathbf{V}$. But it is easily seen that $\nabla^2 g = 0$: it is the Laplacian of $1/R$, the potential of a point charge, which vanishes except at the point $\mathbf{r}$; since the observation site is never inside V , it follows that the second term in the integral in (6) vanishes. Next we apply Gauss's Divergence Theorem to the other term:

$$\Delta \mathbf{B}(\mathbf{r}) = \int_V \nabla \cdot (f \nabla g) d^3 \mathbf{s} = \int_{\partial V} f \hat{\mathbf{n}} \cdot \nabla g d^2 \mathbf{s} \quad (7)$$

where $\hat{\mathbf{n}}$ is the outward facing normal to the volume V , and ∂V denotes the surface of V . Suppose now I choose any smooth function $f(\mathbf{s})$ that vanishes on ∂V . We see from (7) that the magnetic anomaly $\Delta \mathbf{B}$ due to a magnetization $\mathbf{m} = \nabla f$ *vanishes identically* outside V .

A Magnetic Example

- Forward problem - predict magnetic anomaly data B from seamount magnetization vector M
- Inverse problem - find magnetization M from measurements B
- *Magnetic annihilator m has no signal in the data*
- Enter Drastic Simplification - $M(s)$ is uniform - *cannot fit the data*
- Regularization $M(s) = U + R(s)$. What do we know about U ? Not much in linear problems like this.
- Limit ambiguity with additional constraints - e.g. size or direction of M , introduce a norm $\|M\|$

In the seamount magnetization example, the notes discussed which of the following?

